## Practical 1: Create Database and Insert Records

- **CMD / SQLite3 Commands:-**

```
C:\Users\Admin\OneDrive\Desktop\Journal\DB 1>sqlite3 college.db
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> .database
main: C:\Users\Admin\OneDrive\Desktop\Journal\DB 1\college.db r/w
sqlite> .table
sqlite> CREATE TABLE students ( id INTEGER , name TEXT, age INTEGER, course
TEXT );
sqlite> INSERT INTO students VALUES(101,'palak',20,'cs');
sqlite> INSERT INTO students VALUES(102,'ritu',21,'micro');
sqlite> INSERT INTO students VALUES(103,'palu',19,'cs');
sqlite> INSERT INTO students VALUES(104,'mishri',20,'micro');
sqlite> INSERT INTO students VALUES(105,'palki',22,'cs');
sqlite> SELECT * FROM students;
```

- **Output:-**

| id | name | age | course |
|---|---|---|---|
| 101 | palak | 20 | cs |
| 102 | ritu | 21 | micro |
| 103 | palu | 19 | cs |
| 104 | mishri | 20 | micro |
| 105 | palki | 22 | cs |

## Practical 2: Backup Complete Database (.dump equivalent)

- **CMD / SQLite3 Commands:-**

  sqlite> .quit

  C:\Users\Admin\OneDrive\Desktop\Journal\DB 1>sqlite3 college.db ".dump" > college_backup.sql

- **Output:-**

college_backup.sql U

C: > Users > Admin > OneDrive > Desktop > Journal > DB 1 > college_backup.sql

```
1   PRAGMA foreign_keys=OFF;
2   BEGIN TRANSACTION;
3   CREATE TABLE students ( id INTEGER , name TEXT, age INTEGER, course TEXT );
4   INSERT INTO students VALUES(101,'palak',20,'cs');
5   INSERT INTO students VALUES(102,'ritu',21,'micro');
6   INSERT INTO students VALUES(103,'palu',19,'cs');
7   INSERT INTO students VALUES(104,'mishri',20,'micro');
8   INSERT INTO students VALUES(105,'palki',22,'cs');
9   COMMIT;
10
```

## Practical 3: Backup Only Database Structure

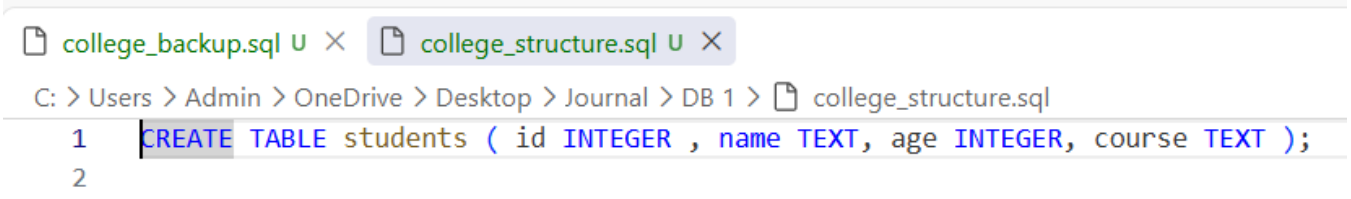
- **CMD / SQLite3 Commands:-**

```
C:\Users\Admin\OneDrive\Desktop\Journal\DB 1>sqlite3 college.db ".dump" >
college_backup.sql

C:\Users\Admin\OneDrive\Desktop\Journal\DB 1>sqlite3 college.db
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> .schema
CREATE TABLE students ( id INTEGER , name TEXT, age INTEGER, course TEXT );
sqlite> .quit

C:\Users\Admin\OneDrive\Desktop\Journal\DB 1>sqlite3 college.db ".schema" >
college_structure.sql
```

- **Output:-**



## Practical 4: Backup Only Data

- **CMD / SQLite3 Commands :-**

```
C:\Users\Admin\OneDrive\Desktop\Journal\DB 1>sqlite3 college.db
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> .mode insert
sqlite> .output college_data.sql
sqlite> .output stdout
sqlite> .quit

C:\Users\Admin\OneDrive\Desktop\Journal\DB 1>sqlite3 college.db "SELECT * FROM
students;" > cig1_backup.sql
```

- **Output:-**

college_backup.sql U × college_data.sql U × cig1_backup.sql U ×

C: > Users > Admin > OneDrive > Desktop > Journal > DB 1 > cig1_backup.sql

```
1  101|palak|20|cs
2  102|ritu|21|micro
3  103|palu|19|cs
4  104|mishri|20|micro
5  105|palki|22|cs
6
```

## Practical 5: Restore Database from Backup

- **CMD / SQLite3 Commands:-**

```
C:\Users\Admin\OneDrive\Desktop\Journal\DB 1>sqlite3 restored_college.db
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> .tables
students
sqlite> SELECT * FROM students;
```

- **Output :-**

| id | name | age | course |
|---|---|---|---|
| 101 | palak | 20 | cs |
| 102 | ritu | 21 | micro |
| 103 | palu | 19 | cs |
| 104 | mishri | 20 | micro |
| 105 | palki | 22 | cs |

## Practical 6: Export Complete Database to Another File

- **CMD / SQLite3 Commands:-**

```
C:\Users\Admin\OneDrive\Desktop\Journal\DB 1>sqlite3 college.db ".backup
college_copy.db"

C:\Users\Admin\OneDrive\Desktop\Journal\DB 1>sqlite3 college_copy.db
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> .tables
students
sqlite> SELECT * FROM students;
```

- **Output:-**

| id | name | age | course |
|---|---|---|---|
| 101 | palak | 20 | cs |
| 102 | ritu | 21 | micro |
| 103 | palu | 19 | cs |
| 104 | mishri | 20 | micro |
| 105 | palki | 22 | cs |

## Practical 7: Display All Tables

- **CMD / SQLite3 Commands :-**

```
C:\Users\Admin\OneDrive\Desktop\Journal\DB 1>sqlite3 college.db
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> .table
```

- **Output:-**

```
sqlite> .table
students
```

## Practical 8: Display Database Schema

- **CMD / SQLite3 Commands :-**

  C:\Users\Admin\OneDrive\Desktop\Journal\DB 1>sqlite3 college.db
  SQLite version 3.53.4 2026-07-24 19:02:57
  Enter ".help" for usage hints.
  sqlite> .schema

  sqlite> .schema students

- **Output:-**

```
sqlite> .schema
CREATE TABLE students ( id INTEGER , name TEXT, age INTEGER, course TEXT );
sqlite> .schema students
CREATE TABLE students ( id INTEGER , name TEXT, age INTEGER, course TEXT );
```

## Practical 9: Count Total Records

- **CMD / SQLite3 Commands:-**

```
C:\Users\Admin\OneDrive\Desktop\Journal\DB 1>sqlite3 college.db
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> SELECT COUNT(*) FROM students;

sqlite> SELECT COUNT(*) AS total_records FROM students;
```

- **Output:-**

```
sqlite> SELECT COUNT(*) FROM students;
┌──────────┐
│ COUNT(*) │
├──────────┤
│        5 │
└──────────┘
sqlite> SELECT COUNT(*) AS total_records FROM students;
┌───────────────┐
│ total_records │
├───────────────┤
│             5 │
└───────────────┘
```